CMR UNIVERSITY

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VALUE ADDED COURSE ON AI/ ML - TASK 2

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COURSE: BCA (GENERAL)

SEMESTER: 5TH / YEAR: III

SUBJECT: ARTIFICIAL INTELLIGENCE

EXPERIMENT 2: House Price Prediction using Kaggle Boston/California Housing Dataset

AIM:

To load and preprocess the California Housing dataset, engineer features, scale the data, train a Ridge Regression model, and evaluate house price prediction using **MAE, RMSE, and R^2 Score**.

SYSTEM REQUIREMENTS:

Hardware Requirements:

▫ Processor: Intel Core i3 or equivalent (minimum)

▫ RAM: 4 GB or higher

▫ Internet Connection: Required (to download the dataset via URL in Google Colab)

Software Requirements:

▫ Operating System: Windows 10/11, macOS, or Linux

▫ Platform: Google Colab / Jupyter Notebook

▫ Environment: Python 3.x

▫ Required Libraries: pandas, numpy, scikit-learn

PROCEDURE:

1. Dataset Acquisition: Load the California Housing benchmark dataset into a Pandas DataFrame.
2. Data Inspection & Cleaning: Check and handle missing values using median imputation.
3. Feature Preprocessing & Engineering: Separate features and target, and create ratio features such as RoomsPerHousehold and BedroomsPerRoom.
4. Dataset Splitting: Split the dataset into 80% training and 20% testing sets.
5. Feature Scaling: Apply StandardScaler to scale the training and testing features.
6. Model Training & Evaluation: Train a Ridge Regression model and evaluate it using MAE, RMSE, and R^2 Score.

PROGRAM:

```
import pandas as pd
import numpy as np
from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import Ridge
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# Step 1: Load Dataset
raw_data = fetch_california_housing(as_frame=True)
df = raw_data.frame

print("--- Step 1: Dataset Loaded Successfully ---")
print(f"Dataset Shape: {df.shape[0]} rows, {df.shape[1]} columns\n")
print(df[['MedInc', 'HouseAge', 'AveRooms', 'AveBedrms', 'MedHouseVal']].head())
print("\n" + "=" * 60 + "\n")

# Step 2: Feature Engineering & Preprocessing
df['RoomsPerHousehold'] = df['AveRooms'] / df['AveOccup']
df['BedroomsPerRoom'] = df['AveBedrms'] / df['AveRooms']

df.fillna(df.median(), inplace=True)

X = df.drop(columns=['MedHouseVal'])
y = df['MedHouseVal']

# Step 3: Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Step 4: Feature Scaling
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

```
# Step 5: Model Training
model = Ridge(alpha=1.0)
model.fit(X_train_scaled, y_train)

# Step 6: Prediction & Evaluation
y_pred = model.predict(X_test_scaled)

mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)

print("--- Step 2: Model Performance on Benchmark Dataset ---")
print(f"Mean Absolute Error (MAE) : ${mae * 100000:.2f}")
print(f"Root Mean Squared Error (RMSE): ${rmse * 100000:.2f}")
print(f"R-squared (R2 Score) : {r2:.4f}\n")

# Step 7: Top Features
feature_weights = pd.Series(model.coef_, index=X.columns)

print("--- Top 3 Positive Predictors (Increasing Value) ---")
print(feature_weights.sort_values(ascending=False).head(3))

print("\n--- Top 3 Negative Predictors (Decreasing Value) ---")
print(feature_weights.sort_values().head(3))
```

OUTPUT:

```
--- Step 1: Dataset Loaded Successfully ---
Dataset Shape: 20640 rows, 9 columns

   MedInc  HouseAge  AveRooms  AveBedrms  MedHouseVal
0  8.3252      41.0  6.984127   1.023810        4.526
1  8.3014      21.0  6.238137   0.971880        3.585
2  7.2574      52.0  8.288136   1.073446        3.521
3  5.6431      52.0  5.817352   1.073059        3.413
4  3.8462      52.0  6.281853   1.081081        3.422

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--- Step 2: Model Performance on Benchmark Dataset ---
Mean Absolute Error (MAE) : $48616.43
Root Mean Squared Error (RMSE): $67535.59
R-squared (R2 Score) : 0.6519

--- Top 3 Positive Predictors (Increasing Value) ---
MedInc               0.790220
RoomsPerHousehold    0.519349
BedroomsPerRoom      0.241390
dtype: float64

--- Top 3 Negative Predictors (Decreasing Value) ---
Latitude    -0.885495
Longitude   -0.833668
AveBedrms   -0.225257
dtype: float64
```

RESULT:

The California Housing dataset was successfully loaded, pre-processed, and used to train a Ridge Regression model. The model predicted house prices and was evaluated using MAE, RMSE, and R^2 Score.